

## Quiz 4 Solution: Applied Natural Language Processing DATA 441/641

1. What makes GPT an **autoregressive** language model?
  - (a) **It models the probability of the next token based solely on previously generated tokens**
  - (b) It generates outputs using a masked sequence-to-sequence loss
  - (c) It processes both past and future tokens during prediction
  - (d) It predicts the current token using a learned embedding of all tokens
2. What would most likely happen if BERT's **segment embeddings** were removed during fine-tuning?
  - (a) Token classification would improve due to fewer learned parameters
  - (b) **The model would struggle to distinguish sentence pairs in tasks like NLI or QA**
  - (c) Attention heads would stop computing positional relationships
  - (d) BERT would automatically compensate using the [CLS] embedding
3. How do **Siamese networks** in NLP ensure consistent learning for sentence similarity?
  - (a) They use different networks to encode each input for maximizing feature diversity
  - (b) **They train using shared weights to map sentences into the same vector space, optimizing a contrastive loss**
  - (c) They replace all attention with convolutional layers for computational efficiency
  - (d) They alternate encoding and decoding phases using variational inference
4. You want to quickly adapt a pretrained language model to a new classification task with only 5 labeled examples. What paradigm best suits this?
  - (a) Learning sentence-level sentiment using millions of examples
  - (b) Generating synthetic tokens to extend pretraining corpora
  - (c) **Using few-shot prompting to classify with minimal supervision**
  - (d) Pretraining a language model on a small vocabulary with constrained syntax
5. Which statement best reflects the use of **prompt engineering** in ChatGPT-style models?
  - (a) **It crafts input phrasing to steer model behavior without additional training**
  - (b) It defines the model's hidden weights through a supervised loss
  - (c) It optimizes the architecture for fixed-sequence generation
  - (d) It uses reinforcement learning to select which tokens to mask in generation
6. What role do **positional, segment, and [CLS] embeddings** play in BERT's input layer?
  - (a) They are concatenated and then passed to the encoder without normalization
  - (b) Only positional embeddings are used for sequence-level prediction
  - (c) They are computed dynamically during training from token co-occurrence patterns
  - (d) **They are summed together to provide token identity, position, and sentence separation context**

7. Which metric would be **least appropriate** for evaluating the fluency and relevance of a sequence-to-sequence model output?
  - (a) **Classification accuracy**
  - (b) ROUGE-L
  - (c) BLEU
  - (d) Cross-entropy loss
8. What is the **primary function** of the attention mechanism in sequence-to-sequence transformers?
  - (a) It transforms positional embeddings into token scores
  - (b) It amplifies signal strength for long sequences during decoding
  - (c) **It learns which input tokens to focus on when generating each output token**
  - (d) It computes dynamic embedding gradients at every time step
9. Why does GPT **restrict attention to previous tokens only** during inference?
  - (a) It follows a reinforcement learning-based training schedule
  - (b) **It matches the pretraining objective where the next token depends only on history**
  - (c) It simplifies embedding computation by avoiding self-attention
  - (d) It uses a bidirectional decoder where such restriction is inherent
10. In **reinforcement learning**, what is the role of the **reward signal**?
  - (a) It determines how often the model updates its architecture
  - (b) It selects which features to use during training
  - (c) **It provides feedback that helps the agent learn which actions are desirable**
  - (d) It encodes the model weights directly into a reward vector